

Discrete Mathematics
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Part I

Module Content

13 Introduction to Discrete Mathematics

13.1 Arrangements and Permutations

13.1.1 Arrangements

Definition 13.1.1.1: An arrangement is a sequence of items in which the order of the items in the sequence matters.

Definition 13.1.1.2: The length of an arrangement is the number of items in the sequence.

Standard Problem 1: How many arrangements of length k can be made from elements of a set S of size $|S| = n$?

Solution: n^k as arrangement has k places, each of which can be occupied by any one of n entries.

Example 13.1.1.1: How many four-letter “words” can be made from the standard A-Z alphabet?

Solution: $26^4 = 456,796$ words.

Standard Problem 2: How many arrangements of length k with no repeats can be made from elements of a set S of size $|S| = n$?

Solution:

$$n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{(n-k)!}, \quad k \leq n \quad (1)$$

As we look at the available choices for each successive item in the list, the number of options is reduced by one each time.

Definition 13.1.1.3: The multiplication principle states that, for an arrangement of length k , if for $i \in \{1, 2, \dots, k\}$ there are n_i choices for the n^{th} item, regardless of previous choices, then the number of possible arrangements is $n_1 n_2 \dots n_k$.

13.1.2 Permutations

Definition 13.1.2.1: A permutation on a finite set S is an arrangement of length $|S|$ that uses each item from S exactly once.

Proposition 13.1.2.1: If $|S| = n$, then there are $n!$ permutations on S .

Proof: This is Standard Problem 2 with $n = k$.

□

Definition 13.1.2.2: We generalise this concept as, for a non-negative integer r , an r -permutation on S is an arrangement of r distinct elements from S . We write $P(n, r)$ for the number of r -permutations on a set of size n , which, by Standard Problem 2, we have,

$$P(n, r) = n(n-1)\dots(n-r+1) = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n \quad (2)$$

13.1.3 Counting Examples

Definition 13.1.3.1: The addition principle states that, for two disjoint, finite sets A and B . Then,

$$A \cap B = \emptyset \quad |A \cup B| = |A| + |B| \quad (3)$$

More generally, if $A_1, A_2, \dots, A_n$ are pairwise disjoint, then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n| \quad (4)$$

13.2 Combinations

Definition 13.2.0.1: A combination from a set of objects is an unordered selection of objects of that set.

Standard Problem 3: How many combinations without repetition do we have of a given size k from a set of size n ?

Solution: First consider that there are $P(n, k) = \frac{n!}{(n-k)!}$ arrangements without repetition. These are ordered lists, sorting them into groups of lists using the same letters will give groups of size $k!$. This means that the number of combinations is

$$\frac{P(n, k)}{k!} = \frac{n!}{k!(n-k)!} \quad (5)$$

Using the binomial coefficient notation, this means the number of combinations is $\binom{n}{k}$.

Example 13.2.0.1: How many ways are there of selecting ten different letters of the alphabet?

Solution: $\binom{26}{10} = \frac{26!}{10!16!}$.

Proposition 13.2.0.1: $\binom{n}{k} = \binom{n}{n-k}$

Proof: This can be seen from the formula for the binomial. □

Proposition 13.2.0.2: Pascal's formula, for $1 \leq k \leq (n-1)$, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Proof: Proof excluded. □

Theorem 13.2.0.1: The binomial theorem, if n is a positive integer and $a, b \in \mathbb{R}$, then

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad (6)$$

Proof: The expression $(a+b)(a+b)\dots(a+b)$ when multiplied out, gives various terms of the form $a^k b^{n-k}$. For fixed k , the number of terms $a^k b^{n-k}$ is $\binom{n}{k}$ since one chooses the k brackets in the product form from which to choose the factor a rather than b . □

13.3 Arrangements and Combinations with Repetitions

13.3.1 Arrangements with Repetitions

Standard Problem 4: How many arrangements are there of a list of objects in which some objects are repeated?

Solution: **Theorem 13.3.1.1:** Given a list of n objects of r different types, in which objects of type i appear n_i times, the number of arrangements is

$$P(n; n_1, n_2, \dots, n_r) = \frac{n!}{n_1! n_2! \dots n_r!} \quad (7)$$

Proof: Proof omitted. □

Example 13.3.1.1: How many arrangements of REARRANGE are there, and of those, how many of the arrangements have at least two R's together?

Solution: The letters to arrange are RRR AA EE G and N. Applying Theorem 13.3.1.1 with $n = 9$ and $n_1 = 3, n_2 = 2, n_3 = 2, n_4 = 1, n_5 = 1$, we get

$$\frac{9!}{3!2!2!} = 15,120 \quad (8)$$

We can then take all the cases where all the R's are separate. To do this, we arrange all the other letters, giving $\frac{6!}{2!2!}$ arrangements. We then insert the R's into the available gaps, giving 7 possible spots for 3 R's, meaning the number of arrangements are

$$\frac{9!}{3!2!2!} - \binom{7}{3} \left(\frac{6!}{2!2!} \right) = 8,820 \quad (9)$$

13.3.2 Combinations with Repetitions

Standard Problem 5: Given a set of n objects, how many combinations of size k can be chosen if objects may be selected more than once? (How many k -combinations with repetitions from n objects are there?)

Solution: To solve this problem, first consider

Standard Problem 6: How many k -combinations with repetitions from n objects are there in which each object is chosen at least once in the combination?

Solution: **Theorem 13.3.2.1:** The number of k -combinations from n different objects with repetition and each object occurring at least once is $\binom{k-1}{n-1}$ for $k \geq n$.

Proof: This question clearly has a non-zero answer only if $k \geq n$. Since order does not matter in our combinations, we represent each combination with its items in non-decreasing order. Now insert markers into the strings at the points where the numbers change. Because we have specified that each number appears at least once, there must be $(n-1)$ markers occupying $(k-1)$ possible positions in each of these lists.

□

Example 13.3.2.1: Count the 5-combinations with repetitions from $\{1, 2, 3\}$ in which each number appears at least once.

Solution: Using the method above, we get the following combinations:

$$\begin{aligned} 11123 &\leftrightarrow 111|2|3 \\ 11223 &\leftrightarrow 11|22|3 \\ 11233 &\leftrightarrow 11|2|33 \\ 12223 &\leftrightarrow 1|222|3 \\ 12233 &\leftrightarrow 1|22|33 \\ 12333 &\leftrightarrow 1|2|333 \end{aligned}$$

There are 4 possible gaps for 2 markers, so the number of ways of arranging the markers is $\binom{4}{2} = 6$.

We can adapt this solution to solve Standard Problem 5 by allowing the markers to also be placed next to one another, or outside of the item strings. Taking this into account, we get:

Theorem 13.3.2.2: The number of k -combinations from n different objects with repetition is

$$\binom{k+n-1}{n-1} = \binom{k+n-1}{k} \quad (10)$$

Proof: As explained above.

□

13.3.3 The Extended Binomial Coefficients

Proposition 13.3.3.1: The principle of upper negation states that, if n and k are positive integers, then

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k} = (-1)^k \binom{k+n-1}{n-1} \quad (11)$$

Proof:

$$\binom{-n}{k} = \frac{(-n)(-n-1)(-n-2)\dots(-n-k+1)}{k!} \quad (12)$$

Factoring out a factor of -1 ,

$$\binom{-n}{k} = (-1)^k \frac{(n)(n+1)(n+2)\dots(n+k-1)}{k!} \quad (13)$$

Which simplifies to give the claimed result.

□

14 Combinatorics

14.1 The Three Principles

14.1.1 Mathematical Induction

Definition 14.1.1.1: Mathematical induction provides a method for establishing the truth of a sequence of statements, each of which is associated with a non-negative integer. In its simplest form, it gives efficient proofs of identities such as the binomial formula.

Theorem 14.1.1.1: The principle of mathematical induction states that: suppose that $P(n)$ is a sequence of propositions labelled by integers $n \geq n_0$. Suppose that

- a. $P(n_0)$ is true, and
- b. For any $n \geq n_0$, the truth of $P(n)$ implies the truth of $P(n+1)$.

Then, for every $n \geq n_0$, $P(n)$ is true.

Proof: $P(n_0)$ is true by assumption (a). Then by (b), $P(n_0+1)$ is also true. Again, by (b), $P(n_0+2)$ is also true, and so on. This chain of deductions eventually reaches any given $n \geq n_0$, so $P(n)$ is true for all $n \geq n_0$.

□

14.1.2 The Pidgeon-Hole Principle

Definition 14.1.2.1: The pidgeon-hole principle states that if n envelopes are distributed among m pidgeon-holes, and $n \geq m$, then at least one pidgeon-hole contains more than one envelope.

14.1.3 The Inclusion-Exclusion Principle

Proposition 14.1.3.1: We have

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3| \quad (14)$$

Proof: Apply $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$ repeatedly.

□

Theorem 14.1.3.1: The inclusion-exclusion formula states that, for a positive integer n and finite sets $A_1, \dots, A_n$,

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{k=1}^n \left((-1)^{k+1} \times \sum_{i_1 < i_2 < \dots < i_k} |A_{i_1} \cup A_{i_2} \cup \dots \cup A_{i_k}| \right) \quad (15)$$

Proof: A slightly tricky induction on n , with the base case $n = 2$ given by $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. □

Definition 14.1.3.1: A derangement of the numbers $\{1, 2, \dots, n\}$ is a permutation in which no number is in its correct position.

Proposition 14.1.3.2: The number of derangements of $1, 2, \dots, n$ is given by

$$d(n) = \sum_{k=0}^n (-1)^k \frac{n!}{k!} \quad (16)$$

Proof: We subtract from $n!$ the number of non-derangements to get

$$d(n) = n! \sum_{k=0}^n (-1)^k \frac{1}{k!} \quad (17)$$

□

In particular, the proportion of permutations that are derangements is

$$\frac{d(n)}{n!} = \sum_{k=0}^n (-1)^k \frac{1}{k!} \quad (18)$$

This is the power series expansion of e^{-1} , so

$$\lim_{n \rightarrow \infty} \frac{d(n)}{n!} = \frac{1}{e} \quad (19)$$

14.2 Recurrence Relations

14.2.1 Basic Definitions

Definition 14.2.1.1: As defined by (Perepelitsa, 2020, p.1), a recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence for all integers n with $n \geq n_0$, where n_0 is a non-negative integer.

Definition 14.2.1.2: As defined by (Perepelitsa, 2020, p.1), a sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

Definition 14.2.1.3: As defined by (Perepelitsa, 2020, p.1), the initial conditions for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Let $\{a_n\}$ be a sequence indexed by integers $n = 0, 1, 2, \dots$, we are given:

- The initial conditions $a_0, a_1, \dots, a_{r-1}$.
- The recurrence relation

$$a_n = \sum_{i=1}^r c_i a_{n-i} = c_1 a_{n-1} + \dots + c_r a_{n-r}, \quad (n \geq r) \quad (20)$$

where $c_1, \dots, c_r$ are specified constants.

Definition 14.2.1.4: The recurrence relation is linear because no term involves more than a single factor of the a_i , and it is homogeneous because every term involves an a_i .

Definition 14.2.1.5: We seek solutions to the recurrence relation (20) of the form $a_n = x^n$, $x \neq 0$. Substituting this into the relation, we get

$$x^n = \sum_{i=1}^r c_i x^{n-i}, \quad (n \geq r) \quad (21)$$

Dividing through by x^{n-r} , we get

$$x^r - \sum_{i=1}^r c_i x^{r-i} = 0 \quad (22)$$

This is the characteristic equation.

14.2.2 General Method for Solving Homogeneous Linear Recurrence Relations

1. Write down the characteristic equation of the recurrence relation as in (22).
2. Find the equation's roots $\alpha_1, \alpha_2, \dots, \alpha_r$.
3. Then, for each i , $a_n = d(\alpha_i)^n$ satisfies the recurrence relation (20) for any constant d .
4. By linearity, any combination of the $(\alpha_i)^n$ also solves the recurrence relation.
5. Find the general solution to the recurrence relation.

- If the roots are distinct, then the general solution to the recurrence relation is:

$$a_n = g(n) = d_1(\alpha_1)^n + d_2(\alpha_2)^n + \dots + d_r(\alpha_r)^n \quad (23)$$

for arbitrary constants $d_1, d_2, \dots, d_r$.

- If α is a root repeated k times, then in place of the expression (23), we use instead

$$a_n = g(n) = (e_1 + e_2 n + \dots + e_k n^{k-1})(\alpha_1)^n + d_2(\alpha_2)^n + \dots + d_r(\alpha_r)^n \quad (24)$$

for constants $e_1, e_2, \dots, e_k$.

6. Apply the initial conditions, giving r linear equations in the r unknown constants d_i . Solve these to get a solution a_n .

Example 14.2.2.1: Solve $a_n = 3a_{n-1} + 4a_{n-2}$, ($n \geq 2$) where $a_0 = 1$ and $a_1 = 0$.

Solution: The characteristic equation is

$$x^2 - 3x - 4 = 0 \quad \rightarrow \quad (x - 4)(x + 1) = 0 \quad (25)$$

This has distinct roots $x = 4$ and $x = -1$, so the general solution is

$$a_n = d_1(-1)^n + d_2(4)^n \quad (26)$$

Using the initial conditions gives

$$1 = a_0 = d_1 + d_2 \quad \text{and} \quad 0 = a_1 = 4d_2 - d_1$$

Hence $d_1 = \frac{4}{5}$ and $d_2 = \frac{1}{5}$, so the solution is

$$a_n = \frac{1}{5}(4^n) + \frac{4}{5}(-1)^n \quad (27)$$

14.2.3 General Method for Solving Inhomogeneous Linear Recurrence Relations

Redefine the recurrence relation (20) to include an inhomogeneous part $f(n)$,

$$a_n = c_1 a_{n-1} + \dots + c_r a_{n-r} + f(n) \quad (28)$$

$f(n)$	Form of Particular Solution $p(n)$
A	B_0 (constant)
An	$B_0 + B_1 n$ (linear)
An^2	$B_0 + B_1 n + B_2 n^2$ (quadratic)
Aq^n	$B_0 q^n$ (exponential)
Anq^n	$(B_0 + B_1 n)q^n$ (etc...)

Table 1: Forms of the Particular Solution

- Find the general solution to the homogeneous problem $g(n) = a_n$.
- Find a specific particular solution $p(n) = a_n$ to the full inhomogeneous problem using Table 1.
 - If $f(n)$ contains q^n where q is a root of the characteristic equation, then q^n should be replaced by nq^{n-1} .
 - If 1 is a root of the characteristic equation, then $f(n) = An^k$ should be considered as $f(n) = An^k \cdot 1^n$ and so should be treated as above.
- The general solution to the inhomogeneous problem is then the sum of these

$$a_n = g(n) + p(n) \quad (29)$$

- Apply the initial conditions to determine the unknown constants.

Example 14.2.3.1: Solve $a_n = a_{n-1} + 2a_{n-2} - 2n$, ($n \geq 2$) where $a_0 = a_1 = 0$.

Solution: Solving the homogeneous recurrence relation, we get the characteristic equation

$$x^2 - x - 2 = 0 \quad \rightarrow \quad (x - 2)(x + 1) = 0 \quad (30)$$

Which has distinct roots $x = -1$ and $x = 2$, giving the general solution

$$g(n) = d_1(-1)^n + d_2(2)^n \quad (31)$$

The inhomogeneous part is $f(n) = -2n$, so we seek a particular solution of the form

$$p(n) = B_0 + B_1 n \quad (32)$$

Setting $a_n = p(n)$ in the inhomogeneous recurrence relation gives

$$B_1 n + B_0 = (B_1(n - 1) + B_0) + 2(B_1(n - 2) + B_0) - 2n \quad (33)$$

Which we solve to get $B_1 = 1$ and $B_0 = \frac{5}{2}$, meaning the particular solution is

$$p(n) = n + \frac{5}{2} \quad (34)$$

This gives us the general solution to the inhomogeneous problem to be

$$a_n = g(n) + p(n) = d_1(-1)^n + d_2(2)^n + n + \frac{5}{2} \quad (35)$$

Substituting in the initial conditions, we get that

$$a_n = -\frac{1}{2}(-1)^n - 2^n + n + \frac{5}{2} \quad (36)$$

14.3 Generating Functions

Definition 14.3.0.1: Generating functions encode combinatorial information into an analytical form so we can solve counting problems. For example, the ordinary generating function for the sequence $a_0, a_1, \dots$ is

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \quad (37)$$

14.3.1 Counting with Generating Functions

Example 14.3.1.1: How many solutions are there to the equation

$$e_1 + e_2 + \dots + e_n = k \quad (38)$$

where each $e_i \in \{0, 1\}$ and n, k are given, positive integers?

Solution: When constructing our generating function, each e_i is associated with the expression $(x^0 + x^1) = (1 + x)$, the generating function is then the n -fold product,

$$f(x) = (1 + x) \cdot (1 + x) \dots (1 + x) = (1 + x)^n \quad (39)$$

If we expand the generating function as a power series, we get

$$(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k \quad (40)$$

Each term x^k corresponds to a choice of x^{e_1} from the first bracket, x^{e_2} from the second bracket, etc., in which the total power of x is $e_1 + \dots + e_n = k$, i.e., a solution to (38).

Lemma 14.3.1.1: For $n \geq 0$, $n \in \mathbb{Z}$, the generating function for the sequence $1, 1, \dots, 1, 0, 0, \dots$ (first $(n + 1)$ terms are 1) is

$$\sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x} \quad (41)$$

Proof: This is the sum of a geometric series which can be proven directly in the form

$$(1 - x) \sum_{k=0}^n x^k = \sum_{k=0}^n (x^k - x^{k+1}) = 1 - x^{n+1} \quad (42)$$

since the sum telescopes.

□

Lemma 14.3.1.2: For $n \geq 0$, $n \in \mathbb{Z}$, the generating function for the sequence $1, 1, 1, \dots$ is

$$\sum_{k=0}^n x^k = \frac{1}{1 - x} \quad (43)$$

Proof: The sum to infinity makes analytical sense only for $|x| < 1$, in which case we can take $n \rightarrow \infty$ in (41) to get (43).

□

Lemma 14.3.1.3: For $n \geq 0$, $n \in \mathbb{Z}$, the generating function for the sequence $\binom{n+k-1}{k}$, ($k \geq 0$) is

$$\sum_{k=0}^n \binom{n+k-1}{k} x^k = \frac{1}{(1-x)^n} \quad (44)$$

Proof: Recall from the extended form of the binomial theorem that

$$(1-x)^{-n} = \sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k \quad (45)$$

As claimed. □

Lemma 14.3.1.4: For $n \geq 0$, $n \in \mathbb{Z}$, if $f(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g(x) = \sum_{k=0}^{\infty} b_k x^k$ are the generating functions of sequences a_k and b_k respectively, then

$$f(x)g(x) = \sum_{k=0}^{\infty} c_k x^k \quad (46)$$

is the generating function of the sequence $c_k = \sum_{l=0}^k a_l b_{k-l}$.

Proof: This is just collecting terms. □

Example 14.3.1.2: Find the coefficient of x^{17} in the expression $f(x) = (1 + x + x^2 + \dots + x^{10})^5$.

Solution: We use Lemma 14.3.1.2 and Lemma 14.3.1.4 to write

$$f(x) = \left(\frac{1-x^{11}}{1-x} \right)^5 = (1-x)^{-5} (1-x^{11})^5 \quad (47)$$

Considering separately the two terms in the product, we have

$$(1-x^{11})^5 = \sum_{i=0}^5 \binom{5}{i} (-1)^i x^{11i} \quad (48)$$

And

$$(1-x)^{-5} = \sum_{j=0}^{\infty} \binom{4+j}{4} x^j \quad (49)$$

The terms in the product

$$\left(\sum_{i=0}^5 \binom{5}{i} (-1)^i x^{11i} \right) \left(\sum_{j=0}^{\infty} \binom{4+j}{4} x^j \right) \quad (50)$$

that contribute x^{17} are $(i, j) = (0, 17), (1, 6)$, so the required coefficient is

$$\binom{5}{0} \binom{21}{4} - \binom{5}{1} \binom{10}{4} = 5,985 - 1,050 = 4,935 \quad (51)$$

14.3.2 Recurrence Relations and Generating Functions

Example 14.3.2.1: Solve for a_n satisfying the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$, ($n \geq 2$) with initial conditions $a_0 = 1$ and $a_1 = -1$.

Idea: Write the generating function for a_n as $f(x) = \sum_{n=0}^{\infty} a_n x^n$. In this sum, substitute in for a_n using the recurrence relation for those values of n for which it is valid, then tidy up the remaining terms using the initial conditions.

Solution: In this case, the recurrence relation allows us to replace a_n by $a_{n-1} + 2a_{n-2}$ for all $n \geq 2$, so we can write

$$f(x) = 1 - x + \sum_{n=2}^{\infty} a_{n-1} x^n + 2 \sum_{n=2}^{\infty} a_{n-2} x^n \quad (52)$$

We want to relate the two sums to $f(x)$ so we take out factors of x so that the index a_k matches the power x^k each time

$$f(x) = 1 - x + x \sum_{n=2}^{\infty} a_{n-1} x^{n-1} + 2x^2 \sum_{n=2}^{\infty} a_{n-2} x^{n-2} \quad (53)$$

We can then re-label the summation indices to get a_n everywhere

$$f(x) = 1 - x + x \sum_{n=1}^{\infty} a_n x^n + 2x^2 \sum_{n=0}^{\infty} a_n x^n \quad (54)$$

We can see that the last term is $\sum_{n=0}^{\infty} a_n x^n = f(x)$ by definition and the first summation is missing the first term, so we can rewrite the equation as

$$f(x) = 1 - x + x(f(x) - 1) + 2x^2 f(x) \quad (55)$$

Rearranging for $f(x)$, we get that

$$f(x) = \frac{1}{1+x} = 1 - x + x^2 - x^3 \dots \quad (56)$$

So our solution is $a_n = 1$ if n is even, and $a_n = -1$ if n is odd.

14.3.3 Partitions

Definition 14.3.3.1: A partition of a positive integer n is an expression of n as an unordered sum of positive integers, called the parts of the partition. We denote the number of partitions of n as $p(n)$.

Theorem 14.3.3.1: The number of partitions $p(n)$ of n is the coefficient of x^n in the power series expansion of the infinite product

$$\prod_{k=1}^{\infty} (1 - x^k)^{-1} = \sum_{n=0}^{\infty} x^n p(n) \quad (57)$$

Proof: Proof omitted. □

14.3.4 Calculus of Generating Functions

Theorem 14.3.4.1: If a_n is a sequence with generating function $f(x) = \sum_{n=0}^{\infty} a_n x^n$, then the generating function for the sequence $b_n = na_n$ is $xf'(x)$.

Proof:

$$\sum_{n=0}^{\infty} na_n x^n = x \sum_{n=0}^{\infty} a_n n x^{n-1} = x \frac{d}{dx} \left(\sum_{n=0}^{\infty} a_n x^n \right) = x f'(x)$$

□

Theorem 14.3.4.2: If $f(x)$ is the generating function of the sequence a_n , then $\frac{f(x)}{1-x}$ is the generating function of the sequence $s_n = \sum_{k=0}^n a_k$.

Proof: Proof omitted. □

15 Graph Theory

15.1 What is a Graph?

15.1.1 Basic Definitions

Definition 15.1.1.1: A simple graph G is a pair of sets $G = (V, E)$ where each element of E is a 2-element subset of V .

Definition 15.1.1.2: Given a graph $G = (V, E)$, if $|V|$ and $|E|$ are finite, then G is a finite graph.

Definition 15.1.1.3: Two vertices are adjacent if there is an edge between them, that is, $u, v \in V$ are adjacent if $\{u, v\} \in E$.

Definition 15.1.1.4: An edge is incident to a vertex if that vertex is one of the endpoints of the edge.

15.1.2 Simple Graphs

Definition 15.1.2.1: The empty/null graph on m vertices is called N_m where $|V| = m$ but $|E| = 0$.

Definition 15.1.2.2: The complete graph on n is called K_n . Given $|V| = n$, we include an edge between every pair of vertices, so $|E| = \binom{n}{2}$.

Definition 15.1.2.3: The cycle graph on n vertices is called C_n . Given $|V| = n$, the cycle graph is constructed by listing the vertices in some order $v_1, v_2, \dots, v_n$ and then adding the cycle of n edges $v_1v_2, \dots, v_{n-1}v_n, v_nv_1$.

Definition 15.1.2.4: The star graph on n vertices is called S_n . Of the $|V| = n$ vertices, $(n - 1)$ are leaves with a single edge to a central hub vertex.

Definition 15.1.2.5: The wheel graph on n vertices is called W_n . It can be constructed from the star graph S_n by adding a cycle around the leaves of S_n .

Definition 15.1.2.6: The n -cube has 2^n vertices and is denoted Q_n .

15.1.3 Handshaking

Lemma 15.1.3.1: For any graph $G = (V, E)$, we have $\sum_{v \in V} \deg(v) = 2|E|$.

Proof: Each edge contributes 2 to the sum of degrees, one to each vertex it is incident to. □

Theorem 15.1.3.1: In any graph, the number of odd-degree vertices is even.

Proof: Let $|V| = n$ and $|E| = m$, and let the number of vertices of odd degree be k . By Lemma 15.1.3.1, we have that the sum of degrees is even, and furthermore the sum of the degrees of the $(n - k)$ vertices of even degree is a sum of even numbers, and thus is even.

Therefore, the sum of the degrees of the remaining k vertices of odd degree must also be even. As a sum of odd numbers, there must be an even number of terms for the sum to be even, and hence there must be an even number of vertices of odd degree. □

15.1.4 Graph Isomorphism, Complements, and Subgraphs

Definition 15.1.4.1: Two simple graphs G_1 and G_2 are isomorphic, $G_1 \cong G_2$, if there exists a bijective function mapping the vertices from G_1 to the vertices of G_2 which preserves adjacency.

Definition 15.1.4.2: Let $G = (V, E)$ be a simple graph. Then the complement of G is $\overline{G} = (V, F)$ where $F = \{uv : u, v \in V \text{ and } uv \notin E\}$.

Definition 15.1.4.3: Given a graph $G = (V, E)$ and a graph $G_1 = (V_1, E_1)$ such that $V_1 \subseteq V$ and $E_1 \subseteq E$, then G_1 is a subgraph of G .

15.2 Eulerian Graphs

15.2.1 Travelling in Graphs

Definition 15.2.1.1: Given a graph $G = (V, E)$, a walk in G is a sequence of vertices $v_1, v_2, \dots, v_k$ from V together with edges $e_1, e_2, \dots, e_{k-1}$ in E such that $e_i = \{v_i, v_{i+1}\}$ for all $1 \leq i \leq (k-1)$ where the length of the walk is the number of edges. If $v_1 = v_k$, the walk is closed.

Definition 15.2.1.2: A trail is a sequence of vertices $v_1, v_2, \dots, v_k$ from V together with distinct edges $e_1, e_2, \dots, e_{k-1}$ in E such that $e_i = v_i v_{i+1}$. That is, a trail is a walk with no edge repeated. A circuit is a closed trail.

Definition 15.2.1.3: A path is a series of distinct vertices, i.e., a walk which never revisits any previous vertex and a cycle is a closed path.

Definition 15.2.1.4: A Hamilton cycle in a graph G is a cycle which visits every vertex of G . A graph which contains a Hamilton cycle is called Hamiltonian.

Lemma 15.2.1.1: Given vertices $u \neq w$ in a graph G , if there is a walk from u to w in G , then there is a path from u to w in G .

Proof: Let W be a walk in G consisting of the vertices v_i , $1 \leq i \leq k$, $v_1 = u$, $v_k = w$ and edges e_i , $1 \leq i \leq (k-1)$. If $v_1, \dots, v_k$ are all distinct, W is a path.

If not, let $x = v_j$ be the first occurrence in W of a vertex which appears more than once, and let $v_{j'}$ be the latest occurrence of that vertex in W . Let W' be the walk formed from W by removing the vertices $v_{j+1}, \dots, v_{j'}$ and the edges $e_j, \dots, e_{j'-1}$. Then x appears only once in W' and so the number of repeated vertices in W' is strictly smaller than W .

We can then repeat this process starting with the walk W' . As each time we repeat, the number of vertices appearing more than once strictly decreases, and eventually we must have a walk containing no repeated vertices, that is, a path. □

Lemma 15.2.1.2: If every vertex in a finite graph G has degree > 1 , then G contains a cycle.

Proof: Let $|V| = n$. Starting from any vertex u in G , then as $d(u) > 1$, there is some edge $e = uv$ incident to u . Let $u = v_1$, $v = v_2$. We construct a cycle in G as follows: at v_2 , as $d(v_2) > 1$, there must be some edge $e_2 \neq e$ incident to v_2 . There are two possibilities:

- (a) There is an edge from v_i to some already visited v_k with $1 \leq k \leq i$.
- (b) We do not have (a), and there is some edge $e_i = \{v_i, v_{i+1}\}$ to some unvisited new vertex v_{i+1} .

But at some point $i \leq n$, we must have situation (a) occurring - we cannot keep finding new vertices. Then the path $v_k, v_{k+1}, \dots, v_i, v_k$ is a cycle in G as required. □

15.2.2 Connectedness

Definition 15.2.2.1: A graph G is connected if, given any two vertices, there is a path between them.

Definition 15.2.2.2: A connected component of $G = (V, E)$ is a subgraph $G_1 = (V_1, E_1)$ of G such that:

- G_1 is connected.
- For any $u \in V_1$ and any $v \in V$ such that there is a path from u to v in G , then $v \in V_1$.
- For every edge $e = uv \in E$ where $u, v \in V_1$, then $e \in E_1$.

15.2.3 Euler's Theorem

Definition 15.2.3.1: An Euler circuit in a graph G is a circuit which uses each edge of G exactly once. A graph with an Euler circuit is called Eulerian.

Theorem 15.2.3.1: Let $G = (V, E)$ be a finite connected graph with $|V| > 1$. Then G has an Euler circuit $\iff$ all vertices of G have even degree.

Proof: We have already seen why $\implies$, to prove $\impliedby$, we require the following Lemma:

Lemma 15.2.3.1: If a graph $G = (V, E)$ is finite and connected, with $|V| > 1$ and all vertices of even degree, then G is a finite union of edge-disjoint cycles, that is, there are cycles $C_1, C_2, \dots, C_n$ such that the union of the vertices of these cycles is equal to V and the union of edges is equal to E .

Proof: Since G is connected, it has no isolated vertices, so all vertices have degree ≥ 2 , by Lemma 15.2.1.2, there is a cycle C_1 in G . We form a new graph from G by removing the all edges from C_1 . The vertices in C_1 have their degrees decreased by 2 and then discard any which have become isolated. We can then repeat the process until the whole graph has been decomposed into cycles.

□

Proof of Theorem 15.2.3.1 ($\impliedby$): Let G be finite, connected, and with all degrees even. Then, by Lemma 15.2.3.1, G is a finite union of n edge-disjoint cycles. We shall show, by induction on n , that these can be combined into a single Eulerian circuit.

- Base Case: If $n = 1$, the one cycle is the Euler circuit.
- Induction Hypothesis: If G is made up of $\geq (n - 1)$ edge-disjoint cycles, it has an Euler circuit.
- Inductive Step: Now consider a graph G made up of n edge-disjoint cycles,

$$G = C_1 \cup C_2 \cup \dots \cup C_{n-1} \cup C_n \quad (58)$$

And let

$$H = C_1 \cup C_2 \cup \dots \cup C_{n-1} \quad (59)$$

As in the proof of Lemma 15.2.3.1, H has all degrees even. It may not be connected, so consider its connected components, $H_1, H_2, \dots, H_k$. Since each cycle C_i lies within only one connected component H_j , we know that there are at most $(n - 1)$ connected components.

Furthermore, since the union $C_1 \cup C_2 \cup \dots \cup C_{n-1}$ makes up the whole of H , each H_j must be made up by $\leq (n - 1)$ edge-disjoint cycles. So, by the Induction Hypothesis, each connected component H_j has an Euler circuit E_j .

But since G was connected, C_n must visit each H_j , so we can make an Euler circuit for G by following C_n , but when we first visit each H_j , follow E_j all the way round before continuing.

□

15.3 Planar Graphs

15.3.1 The Three Utilities Problem

Definition 15.3.1.1: The three utilities problem asks whether we can connect gas, water, and electricity to three houses without the pipes/wires crossing?

Definition 15.3.1.2: Given a graph $G = (V, E)$ where there exists $V_1, V_2 \subseteq V$ such that $V_1 \cup V_2 = V$, $V_1 \cap V_2 = \emptyset$, and every edge of G is of the form uv where $u \in V_1$ and $v \in V_2$, then G is bipartite.

Definition 15.3.1.3: A graph G is planar if it can be drawn in the plane with no edges crossing. Such a drawing is a planar representation of G .

15.3.2 Faces and Boundaries

Definition 15.3.2.1: A planar representation of a graph G divides the plane into a set of F faces. The boundary of a face is the set of edges around it, considered as a walk in G . For a face $f \in F$, we write $b(f)$ for the length of the boundary of f .

Lemma 15.3.2.1: Planar Handshaking, for any planar representation of a graph $G = (V, E)$, we have

$$\sum_{f \in F} b(f) = 2|E| \quad (60)$$

Proof: Each edge contributes 2 to the sum of boundary lengths, one for each side of the edge. □

Definition 15.3.2.2: A bridge in a graph G is an edge e of G such that removing e from G increases the number of connected components.

Lemma 15.3.2.2: Given any edge e of a graph G , then e is a bridge $\iff e$ is not part of a cycle in G .

Proof: Let e join the vertices u and v of G . We prove the contrapositive - if e is part of a cycle C , then there is a path P between u and v formed by removing e from C . Then e is not a bridge, since given any path between vertices w_1 and w_2 which passes through e , we can traverse P instead of the edge e to find a path between w_1 and w_2 .

As this is possible, removing e would not increase the number of connected components. Conversely, if e is not a bridge, then considering the graph $G' = G - e$, we have u, v are in the same connected component in G' , so there is a path from u to v in G' , and adding e to this path forms a cycle in G . □

Theorem 15.3.2.1: The Jordan Curve Theorem states that, a simple closed curve in the plane divides it into two regions, inside and outside.

Proof: Proof omitted. □

Theorem 15.3.2.2: In a planar representation of a finite graph, e is a bridge $\iff$ both sides of e are on the same face.

Proof: By Lemma 15.3.2.2, if e is not a bridge, it is contained in a cycle, which forms a simple closed curve in the plane, and so by the Jordan Curve Theorem the two sides of e are different regions. □

15.3.3 Euler's Formula

Theorem 15.3.3.1: Euler's Formula, Let $G = (V, E)$ be a planar graph, with the set of faces F , and k connected components. Then

$$|V| - |E| + |F| = k + 1 \quad (61)$$

Proof: By induction on $|E|$.

- Base Case: When $|E| = 0$, G is a null graph, so $|F| = 1$ and $k = |V|$.
- Inductive Hypothesis: Assume Euler's formula is true for any graph with $|E| - 1$ edges.
- Inductive Step: Draw $G = (V, E)$, which has $|E|$ edges, in the plane. Remove one edge e to make G_0 . G_0 has $|V|$ vertices, $|E| - 1$ edges, and, say, $|F_0|$ faces and k_0 components. The induction hypothesis applies for G_0 , so we know that

$$|V| - (|E| - 1) + |F_0| = k_0 + 1 \quad (62)$$

$|F_0|$ and k_0 relate to $|F|$ and k in one of two cases, either:

1. e is a bridge in G . Removing it makes one more component in G_0 , so $k_0 = k + 1$. By Theorem 15.3.2.2, its two sides bound the same face, so $|F_0| = |F|$, and (62) becomes

$$|V| - (|E| - 1) + |F| = (k + 1) + 1 \quad (63)$$

2. e is not a bridge in G . Removing it does not disconnect G , so $k_0 = k$. But by Theorem 15.3.2.2, in G its sides bounded two distinct faces, which now become one. So (62) becomes

$$|V| - (|E| - 1) + (|F| - 1) = (k + 1) + 1 \quad (64)$$

In both cases, we have $|V| - |E| + |F| = k + 1$, so Euler's formula holds for G .

□

15.3.4 Simple Planar Graphs

Theorem 15.3.4.1: If $G = (V, E)$ is a simple planar graph with $|V| \geq 3$, then

$$|E| \leq 3|V| - 6 \quad (65)$$

Proof: Provided G has at least two edges, then every face has $b(f) \geq 3$, so Lemma 15.3.2.1 gives

$$2|E| = \sum_{f \in F} b(f) \geq 3|F| \quad (66)$$

But also,

$$|V| - |E| + |F| = k + 1 \quad (67)$$

So,

$$\begin{aligned} |V| - |E| + \frac{2}{3}|E| &= k + 1 \geq 2 \\ |V| - |E| + \frac{2}{3}|E| &\geq 2 \\ |V| - 2 &\geq \frac{|E|}{3} \end{aligned}$$

Which can be rearranged to give (65) as claimed.

□

Lemma 15.3.4.1: K_5 is not planar.

Proof: For $G = K_5$ we have that $|V| = 5$ and $|E| = 10$ so $|E| \not\leq 3|V| - 6$.

□

Definition 15.3.4.1: The girth g of G is the shortest length of a cycle in G . If G has no cycles, then $g = 0$.

Theorem 15.3.4.2: If G is simple, planar, and with girth $g \neq 0$, then

$$|E| \leq \frac{g}{g-2}(|V| - 2) \quad (68)$$

Proof: Here every face has $b(f) \geq g$, so we get

$$\frac{2|E|}{g} \geq |F| \quad (69)$$

from Lemma 15.3.2.1. Again, using $|V| - |E| + |F| = k + 1$, we get

$$|V| - |E| + \frac{(g-2)}{g}|E| \geq 2 \quad (70)$$

Which can be rearranged to get (68) as claimed. □

Lemma 15.3.4.2: $K_{3,3}$ is not planar.

Proof: Since $K_{3,3}$ is bipartite, it has no cycles of odd length, and so we can see $g = 4$. We have $|V| = 6$ and $|E| = 9$, so

$$\frac{g}{g-2}(|V| - 2) = \frac{4}{2}(6 - 2) = 8 \quad (71)$$

and $|E| = 9 \not\leq 8$. □

Definition 15.3.4.2: A subdivision of G is a graph obtained from G by subdividing edges of G , where subdividing an edge $e = uv$ if G means

- (a) Adding a new vertex w to G .
- (b) Adding edges uw, wv .
- (c) Removing the original edge e .

Theorem 15.3.4.3: From (Kuratowski, 1930), a graph G is planar $\iff G$ contains no subdivision of K_5 or $K_{3,3}$ as a subgraph.

Proof: $\Leftarrow$ follows from Lemma 15.3.4.1 and $\Rightarrow$ from Lemma 15.3.4.2. □

Theorem 15.3.4.4: Every finite simple planar graph has a vertex of degree ≤ 5 .

Proof: Clearly true for $V \leq 6$. For $|V| > 6$, suppose all vertices are of degree ≥ 6 , then by handshaking, $|E| \geq 3|V|$. Then G does not satisfy Theorem 15.3.4.1 and so is not planar. □

15.3.5 Platonic Graphs

Definition 15.3.5.1: A Platonic graph is finite, simple, planar, and connected, with all vertices of the same degree $d \geq 3$, and all faces of the same boundary length, $b \geq 3$.

d	b	 V 	 E 	F	
3	3	4	6	4	tetrahedron
3	4	8	12	6	cube (hexahedron)
4	3	6	12	8	octahedron
3	5	20	30	12	dodecahedron
5	3	12	30	20	icosahedron
from (74)	from (74)	from (73)	from (72)	from (72)	

Table 2: Platonic Graphs

Theorem 15.3.5.1: There are just five Platonic graphs.

Proof: By handshaking,

$$d|V| = 2|E| = b|F| \quad (72)$$

So Euler's formula $|V| - |E| + |F| = k + 1$ becomes

$$|V|(2b - bd + 2d) = 4b \quad (73)$$

But $|V| > 0$ and $b > 0$ so

$$(b - 2)(d - 2) < 4 \quad (74)$$

So we have Table 2.

□

Part II

References

16 Module References

16.1 Module Professors

Introduction to Discrete Mathematics and *Combinatorics* was taught by Professor Mikhail Menshikov.

Graph Theory is taught by Irene Ayuso-Ventura.

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16.2 Cited Works

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